ELECTRICAL AND INSTRUMENTATION CABLE MANAGEMENT PROCEDURE

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1. PURPOSE

The purpose of this procedure is to optimize E&I cable management from engineering phase until its installation at site, to optimize it. Through this management, we could plan the different pulling cables works according to our priorities and schedule at site, reduce the excess of cable, detect the shortage of cable in an early stage of the project so any additional purchase can be arranged soon, and to have our contractor's works better controlled. This shall enable a better control over the certification process as well as the back charges to apply.

This document is part of the **Intecsa Industrial/Initec Energía** Quality Management System and has been prepared taking into consideration the ISO 9001 standard.

2. ROLLS AND RESPONSIBILITIES

Below are established main function and responsibilities of every part involved in the E&I cable management.

2.1 Engineering Department

- Based on project schedule, submit all the necessary documents for a correct pulling cable (cable list, cable drums schedule and distribution layouts drawings) in order to have clear information regarding cable routing, pulling cable lengths and cable drum from where it must come out. The length stated in the cable list already includes an additional percentage in order to absorb any deviations in the measurement it might be since it is a theoretical value measure from the drawings. This additional percentage includes the cable connection. Therefore, no additional cable respect the indicated in the cable list shall be pulled on Site, and if it were necessary, it shall be properly justified. This deviation shall be reflected in the Pulling Cable Laying sheet MI-42, using remarks / deviation cause column, to categorize the deviation. Electrical or Instrumentation supervisor, together with Subcontractor, shall sign the document.

- Official submission of engineering documents as per Project Quality Plan as well as necessary revisions to update the changes / modifications due to detailed engineering development and / or to include changes requested as a result of the progress of the work at site that could result in re-assignation of cable drums or additional purchase of cables. Electrical and Instrumentation Department shall also be informed in case pulling cables lengths were less than expected and an additional surplus is generated and has to be added to the one already established according to engineering criteria.

  Cable drums that shall be considered for new assignation shall be those that have not been used by the subcontractor, whether they are at warehouse or under the subcontractor's possession but without being used. During this process it shall not be allowed to take more cable drums from the warehouse or that the subcontractor uses new cable drums unless expressly authorised by technical office. It is for this reason that re-assignations process must be done as quickly as possible.

- In contact with site, when necessary and requested by Technical Office / E&I supervisor, in order to clarify design doubts and discrepancies between planned pulling cable lengths in the cable list and the estimated to be executed at site.
- Submission of MTOs / Material Requisition for purchase, as a consequence of engineering documents issuance, as well as their different update revisions, to ensure the early detection and procurement of the additional cable material that might be needed in order to not jeopardize construction works continuity.
- Packages units are sometimes included in project scope of work. In this case in which vendors develops detail engineering as well as supplies, our own engineering requirements in terms of deliverables submission (cable list, cable drums schedule and distribution layouts drawings) will be extended to them and it shall be reflected in V.R.D.D (vendor required documents and date) of each MR (material requisition). This shall assure to control and monitor that every pulling cable works comes from the designated cable drum and detect in an early stage of the work any deviation in the theoretical length in order to be communicated to the vendor to remediate this lack of material in the supply.

2.2 Construction / Supervision Department

- Plan in advance, together with the subcontractor, and based on construction priorities and planning work packages, the works to be done.
- Check at site, with the cable drum in situ and before cutting, if the expected pulling cable length corresponds to the one defined by engineering in the cable list. In case of discrepancies between theoretical measure and what finally is going to be pulled, Electrical and Instrumentation Supervision team (otherwise, technical office), shall be the interlocutor with Electrical and Instrumentation Engineering discipline, to clarify and report the existing deviations. Changes shall be documented in as-built drawings and in Pulling Cable laying sheet MI-42 including the root cause of the deviation, so E&I engineering disciplines, in a continuous improvement process, shall be able to use better surplus factors in the cable list.
- Pulling cable works shall be in accordance to the most updated engineering documents issued for construction. Superintendent or E&I Supervisor (otherwise, technical office), shall report to E&I Engineering discipline of any deviation it might be as soon as it´s detected, so cable drums can be re-assigned from Home Office.
- Before and after each pulling cable work, it is necessary a cable length remaining double check in the drum schedule to make sure that matches with what is stated in the pulling cable laying sheet MI-42.
- Pulling cable works shall be done from the specified cable drums according to cable drum schedule and if it were necessary to leave some cable for a later connection with the equipment, this additional cable length shall be according to construction best practices. As a general criteria, a minimum of 3 meters per each side shall be available for connection and a maximum of 5 meters per each side for pulling cable length longer than 200 meters.
- Cable drums shall be returned to warehouse for its control. In case drums are given to subcontractors for its own management, remaining cable in each cable drum shall be checked after every pulling cable work.
- Pulling cable laying sheet MI-42 that comes together with cable drum and is where subcontractor records pulling cable lengths, must be updated and show pulling cable work and lengths of each routing as well as deviations from theoretical length, so a shortage of material shall be detected easily and quickly.

- Pulling cable laying sheet MI-42 shall be signed by Contractor and subcontractor to confirm the information it contains.

2.3 Subcontracts Department

- Subcontractor's scope of work must be defined and delimited.
- To assure the subcontract includes the obligation for the Subcontractor to fulfill this procedure requirements, and in case of breach of the procedure, they could be economically back charged due to the consequences their actions might have.
- To establish maximum cable lengths for end connections if necessary, because the equipment has not been installed yet. As a general criteria, a minimum of 3 meters per each side shall be available for connection and a maximum of 5 meters per each side for pulling cable length longer than 200 meters.
- To assure Unit rates descriptions and methods of measurement are included in the subcontract and reflect the criteria of payment is for net pulling cable lengths according to theoretical distribution layouts documents issued for construction
- To assure the subcontract includes the material balance formula defining a reasonable percentage for "Irrecoverable Scrap".
- To assure the Subcontract includes the obligation to subcontractor to fill the pulling cable laying sheet MI-42, including the actual pulled lengths, and that it is signed by the subcontractor and our E&I supervision.

2.4 Subcontract Administration Department

- To assure subcontract clauses are fulfilled
- Unit rates descriptions and methods of measurement included in the subcontract shall reflect the criteria of payment for net pulling cable lengths according to theoretical distribution layouts documents issued for construction
- For electrical secondary cable and cable trays, engineering discipline shall try to develop as much as possible the design and take the cable to the closest point to the equipment or wherever enough available space is. Nevertheless, this design is highly susceptible to change at site. In this case, subcontractor together with Electrical supervision team / Technical Office shall establish the routing of this cable.

- For instrumentation secondary cable and cable trays, the routing of these cables cannot be fixed during detail engineering development because is highly affected by instrument layout, construction progress, interferences, scaffolding…etc. In this case, subcontractor together with Instrumentation supervision team / Technical Office shall also establish the routing of this cable.
- Subcontractor cannot acts on its own. If this were the case, economical certifications and measurement documents shall be rejected and Contractor will reserve the right to back charge the cable misuse when it is proved that its routing has not been the most optimized.
- To check that economical certifications are based on net pulling cable lengths criteria. Certification could not include lengths resulting from material misuse. These shall be back charged when applicable.
- Check and Control that all excess material / scrap, have been returned; and in case this doesn't occur, or and additional material have been purchased because of misused, it´s back charge to subcontractor..
- Once deviations have been detected between theoretical length from drawings and actual length already pulled, it is necessary to categorized them and reflect in the pulling cable laying sheet MI-42:
    - Estimation errors (cable tag and cable drum specified by engineering have been properly used, cable routing has been the one designed by engineering but cable length differs from the one stated in the cable list).
    - Different scope: New cables have appeared during the project that were not considered in the estimation.
    - Installation errors, misuse, loss, excess cable for connection: to be back charged to the subcontractor.

2.5 Procurement Department

- Purchase the necessary material in accordance to engineering MTOs / Material Requisition documents. Engineering shall be up to date on possible shortage of cable material and therefore shall issue a new MTO / Material Requisition for purchase to allow the additional material to be purchased. Purchase being done locally market or from Home Office shall depend on the circumstances (deadlines,

stock at site, etc). In any case, the purchase order shall be registered in **Intecsa Industrial/Initec Energía** IT systems.

- Lighting and earthing cables are not included in cable list. So it´s possible it would be necessary to purchase additional material as per site request. In this cases, E&I Supervision / Technical Office, shall communicate to E&I Engineering Discipline new requirements in order to update MTO and MR to proceed with material procurement.
- Request in purchase conditions to vendors and double check that cable drums come with their corresponding identification tag and marked with cable linear meters, and not at end and at the beginning of the cable drum; marking shall be on each one or five meter, if not, subcontractor will cut blindly. This requirement has to be clearly shown in the material requisition / purchase order, and its fulfilment shall be verified before the release for delivery of the material by inspection teams. This requirement has to be also applicable for local purchase orders. This may help to verify remaining material in cable drums when it´s returned to warehouse as well as fill cable laying sheet MI-42 after every laying cable work.
- Cables drums shall be delivered on time at site. The delay might cause using upgrade cables for some works even when they weren´t specified by engineering for that use in order to catch the schedule plan time or to finish an urgent one (trench to be backfilled)
- Inspection team shall assure the material is released for delivery as per commercial conditions and without damaged.

2.6 Warehouse / Material Management (SIE) Department

- To check the drums have been delivered as per purchase order requirements. During the reception is checked:
    - Visually, to check the linear meters indicated and engraved well on the cable sheath.
    - Testing, make the required test to megger and apply continuity on the cable drums, to check if any issue is wrong before keeping the drum in the warehouse.
- To have located all cable drums at the warehouse and provide them to subcontractors according to exit vouchers. Pulling cable laying sheet MI-42 shall be

delivered together with cable drum and shall indicate pulling cables works that have to come out from each of them. This document must include theoretical lengths according to cable list and drum schedule information and specific boxes to be filled in the cut lengths. The document shall be signed by the Superintendent or E&I Supervisor and subcontractor.

Once the cable pulling works from a specific Cable Drum have been finished and the pulling cable laying sheet MI-42 has been properly filled and signed by E&I Supervision and Subcontractor, the document shall be internally distributed to all concerned department (E&I supervision, technical office, cost control, subcontract administrator and E&I engineering disciplines in Home Office).

- When site work circumstances and warehouse physical location allows it, and material has to exit and enter several times from the warehouse, they shall be responsible for checking remaining cable lengths in the drum, so once it is delivered again (maybe to other subcontractor) there are no incidents with the quantities delivered.
- Material Management team (SIE) shall develop constructability studies and analyze work front availability according to construction priorities, planning, MTO's, purchased cables and those already delivered at site.

2.7 Subcontractors

- Planning in advance the works together with Superintendent or E&I supervision, according to Project needs and priorities.
- Double check at site that the actual lengths to be cut are the ones indicated in the cable list and shall inform to Superintendent or E&I supervision of any deviation it might be.
- Efficient use of the material delivered.
- Return any excess of material or scrap to warehouse.
- Issue pulling cable laying sheet MI-42 filled in and signed by Subcontractor and our own Superintendent or E&I supervision.
- Subcontractor shall have his own yard, just to preserve properly the number of drums which always frequently move from location to other.

- Certification process shall be based on unit rates descriptions and methods of measurement included in the subcontract that shall reflect the criteria of payment is for net pulling cable lengths according to theoretical distribution drawing issued for construction.
- To prepare and submit the economical certifications based on net pulling cable lengths criteria. Certification cannot include lengths resulting from material misuse. These shall be back charged when applicable

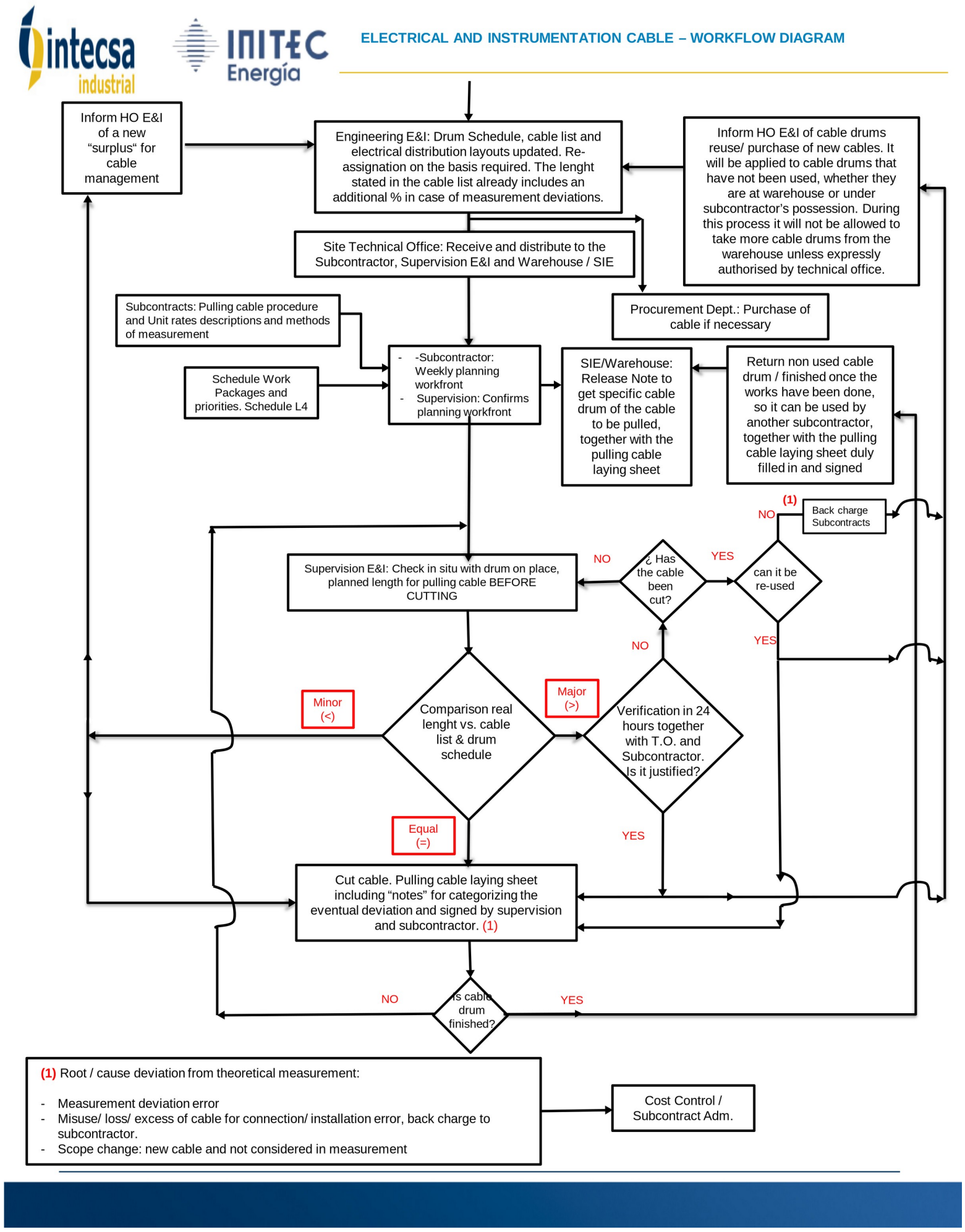
2.8 Technical Office

- Support on site construction works, when necessary, to solve discrepancies between what has been designed and what is going to be pulled.

2.9 Planning Department

- A “work packages” (WBS) planning shall be prepared in advance according to project needs and priorities. This plan must be regularly updated to reflect actual progress and considering that priorities might change.
- Project must be focus and designed based on these “work packages” so for the first item we are going to install, the design is ready, equipment to be cable connected is at site and installed and cable to be pulled available.
- Start dates and duration for different work areas shall be extracted from project baseline Schedule and included in the annex of the subcontract as a time frame for subcontractor to perform the job as well as a base of level IV schedule development. The fulfilment of this specific schedule shall allow the completion of project works as per overall completion dates.

2.10 Quality Department

- Site Quality Control Manager and his team will control and monitor the correct fulfilment of this document.

Attachment I. Electrical and Instrumentation Cable Workflow Diagram